

Kerberos

Kerberos

- The mythical three headed dog guards the gates to the underworld.
- Ticket authentication
- After authentication Kerberos Issues tickets
- Tickets are then used by the user when accessing resources.

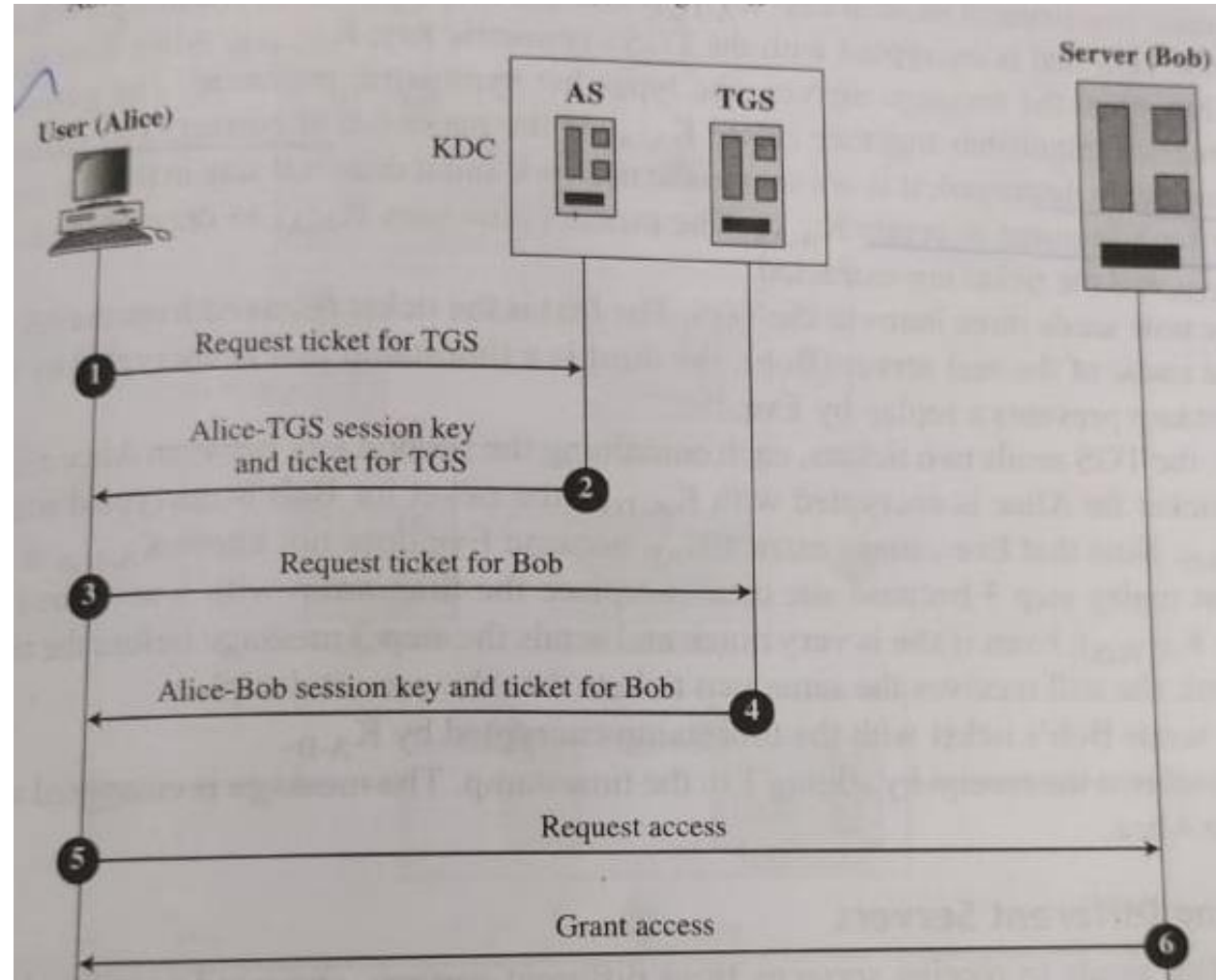
Benefits:

1. Single sign-In.
2. Protects log-on credentials.

Kerberos cntd..

- Kerberos is an authentication protocol, and at the same time a KDC, that has become very popular.
- Three servers are involved in the Kerberos protocol; authentication server (AS), a ticket-granting server (TGS), and a real (data) server that provides services to others.
- AS is the KDC in the Kerberos protocol. Each user registers with the AS and is granted a user identity and a password. The AS has a database with these identities and the corresponding passwords. The AS verifies the user, issues a session key to be used between Alice and the TGS, and sends a ticket for the TGS.
- TGS Issues a ticket for the real server (Bob). It also provides the session key (K_{ab}) between Alice and Bob. Kerberos has separated user verification from the Issuing of tickets. In this way, though Alice verifies her ID just once with the AS, she can contact the TGS multiple times to obtain tickets for different real servers.
- Real Serve(Bob) provides services for the user (Alice). Kerberos designed for a client-server program, such as FTP.

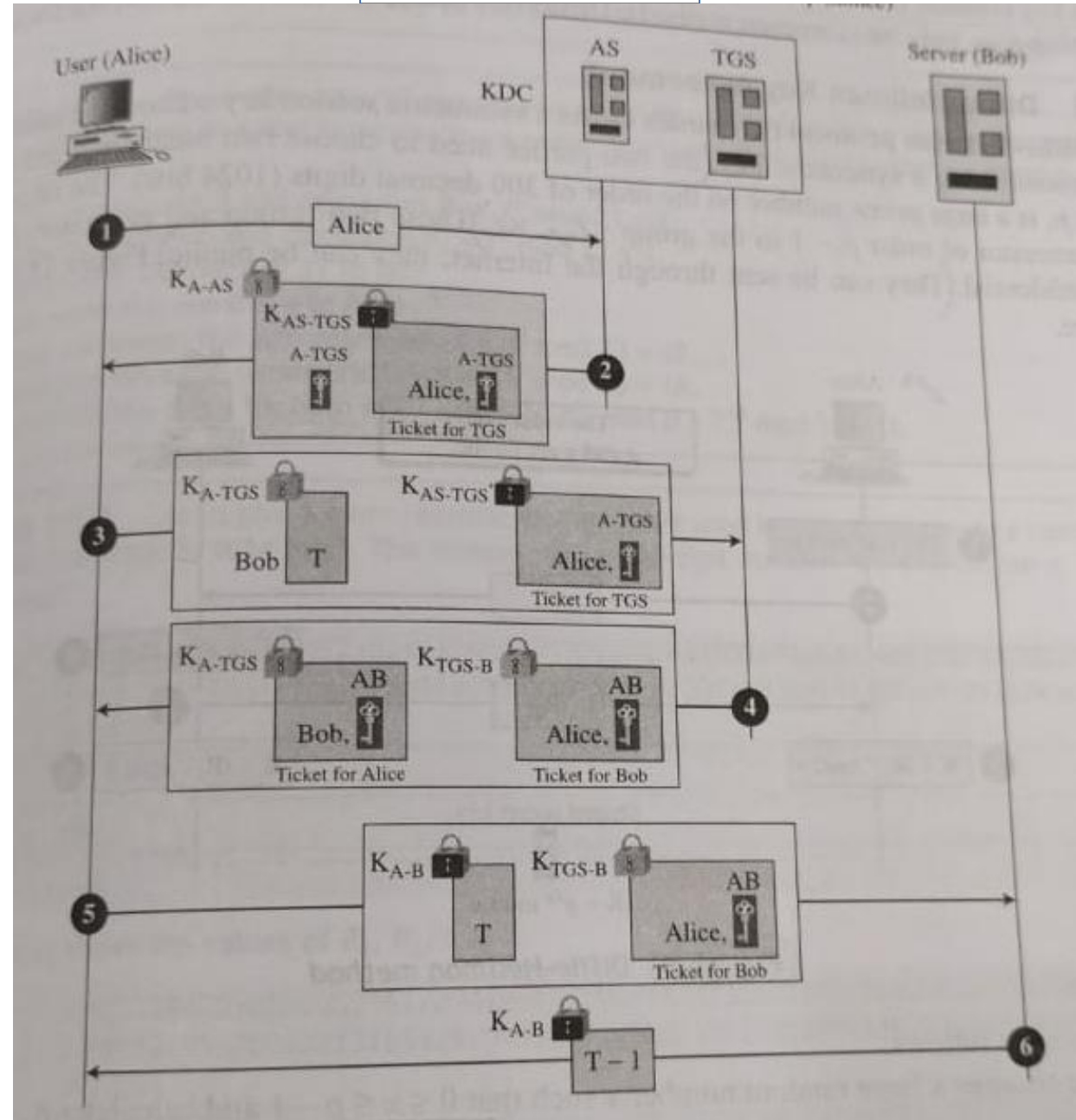
Kerberos cntd..



Three servers

Kerberos cntd..

Kerberos Example



Operation

Step1,2

2.2 Operation

Client process (Alice) can access a process running on the real server (Bob) in six steps, as shown in

15.8

Alice sends her request to the AS in plain text using her registered identity.

The AS sends a message encrypted with Alice's permanent symmetric key, K_{A-AS} . The message contains two items: a session key, K_{A-TGS} , that is used by Alice to contact the TGS, and a ticket for the TGS that is encrypted with the TGS symmetric key, K_{AS-TGS} . Alice does not know K_{A-AS} , but when the message arrives, she types her symmetric password. The password and the appropriate algorithm together create K_{A-AS} if the password is correct. The password is then immediately destroyed; it is not sent to the network and it does not stay in the terminal. It is used only for a moment to create K_{A-AS} . The process now uses K_{A-AS} to decrypt the message sent. K_{A-TGS} and the ticket are extracted.

The first is the ticket received from the AS. The second

Operation

Step 3,4

K_{A-TGS} and the ticket are extracted. Alice now sends three items to the TGS. The first is the ticket received from the AS. The second is the name of the real server (Bob), the third is a timestamp that is encrypted by K_{A-TGS} . The timestamp prevents a replay by Eve.

Now, the TGS sends two tickets, each containing the session key between Alice and Bob, K_{A-B} . The ticket for Alice is encrypted with K_{A-TGS} ; the ticket for Bob is encrypted with Bob's key, K_{TGS-B} . Note that Eve cannot extract K_{AB} because Eve does not know K_{A-TGS} or K_{TGS-B} . She cannot replay step 3 because she cannot replace the timestamp with a new one (she does not know K_{A-TGS}). Even if she is very quick and sends the step 3 message before the timestamp has expired, she still receives the same two tickets that she cannot decipher.

Operation

Step 5,6

Alice sends Bob's ticket with the timestamp encrypted by K_{A-B} .
Bob confirms the receipt by adding 1 to the timestamp. The message is encrypted with K_{A-B} and sent to Alice.

Difference between version 4 and 5 of kerberos

- Version 5 has a longer ticket lifetime.
- It allows tickets to be renewed.
- It can accept any symmetric-key algorithm
- It uses a different protocol for describing data types
- It has more overhead than version 4.